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actors are increasingly recognized as factors influencing surgical outcomes in orthopedic surgery. Smoking, for example, is a well-documented risk factor for poor wound healing, infection, and implant-related complications⁽⁸⁾. Similarly, hypoalbuminemia reflects compromised nutritional and inflammatory status and has been linked to adverse postoperative outcomes, while elevated body mass index (BMI) may contribute to mechanical overload and altered biomechanics around the ankle joint⁽⁹⁾. Despite their biological plausibility and clinical relevance, these factors are infrequently incorporated into comparative analyses of ankle arthrodesis and arthroplasty, particularly within time-to-event frameworks.

Propensity score matching (PSM) offers a robust method for minimizing baseline confounding in observational studies by balancing measured covariates between treatment groups. When combined with survival analysis using Cox proportional hazards modeling, this approach allows for an adjusted comparison of the risk of implant failure over time while accounting for both treatment selection bias and independent risk factors⁽¹⁰⁾. To date, large-scale studies integrating PSM with multivariable time-to-event analysis to evaluate implant failure following ankle arthrodesis versus total ankle arthroplasty remain limited.

Therefore, the purpose of this study is to compare the risk of implant failure between ankle arthrodesis and total ankle arthroplasty in a large multicenter retrospective cohort, using propensity score matching to balance demographic and clinical covariates, followed by an adjusted Cox proportional hazards model accounting for metabolic and nutritional factors, including hemoglobin A1c, serum albumin, and body mass index.

Methods

Study design and data source

As with all analyses based on administrative and electronic health record data, the TriNetX platform does not provide detailed information on implant brand, fixation method, surgical approach, deformity severity, or surgeon volume and experience.

Study population

Adult patients aged 18 to 100 years with a diagnosis of ankle osteoarthritis were eligible for inclusion. Osteoarthritis was defined using ICD-10-CM codes for primary osteoarthritis of the ankle and foot (M19.07), post-traumatic osteoarthritis of the ankle and foot (M19.17), or secondary osteoarthritis of the ankle and foot (M19.27).

Patients were included if they underwent either ankle arthrodesis or total ankle arthroplasty between January 1, 2012, and December 31, 2023. The date of the qualifying surgical procedure was defined as the index date.

Cohort definitions

Two mutually exclusive cohorts were defined based on procedural codes:

Cohort 1: Ankle arthrodesis – Patients with qualifying osteoarthritis diagnoses who underwent ankle arthrodesis, identified using CPT and SNOMED procedure codes for ankle arthrodesis. Patients with any record of ankle arthroplasty were excluded from this cohort.

Cohort 2: Total ankle arthroplasty – Patients with qualifying osteoarthritis diagnoses who underwent ankle arthroplasty, identified using CPT codes for total ankle replacement. Patients with any record of ankle arthrodesis were excluded from this cohort.

Patients with neoplastic diagnoses (ICD-10-CM C00-D49) were excluded from both cohorts.

Outcome definition

The primary outcome was implant failure, defined using ICD-10 codes T84.0, T84.1, and T84.2 (complications of internal orthopedic prosthetic devices, implants, or grafts). Importantly, this code represents a composite, coding-based outcome and does not distinguish among specific failure mechanisms. In ankle arthrodesis, implant failure codes may reflect nonunion-related hardware failure, symptomatic or broken fixation devices, or infection. In contrast, in total ankle arthroplasty, the same codes more commonly capture prosthesis-related complications such as loosening, wear, or infection. Therefore, 'implant failure' should be interpreted as a heterogeneous administrative outcome rather than a uniform biological or mechanical failure across procedures.

Covariates

Baseline characteristics assessed at or prior to the index date included demographic variables (age, sex, race, and

nsion, obesity, nicotine dependence, monary disease, Charcot arthropathy, d prior infection), prior procedures, cluding systemic corticosteroids and anabolic steroids), and laboratory parameters.

Variables of interest included anthropometric measurements and laboratory parameters. Anthropometric assessment included BMI. Laboratory variables were analyzed using a complete-case approach. Hemoglobin A1c and serum albumin were available for a subset of patients only (approximately 14% and 23% of the matched cohort, respectively), and no data imputation was performed, as this is not supported by the TriNetX platform.

Propensity score matching

To minimize confounding due to baseline differences between cohorts, PSM was performed using all listed baseline characteristics. Propensity scores were estimated using logistic regression, with cohort assignment (ankle arthrodesis versus total ankle arthroplasty) as the dependent variable.

Propensity score matching was performed using nearest-neighbor matching in a 1:1 ratio, without an explicit caliper width, consistent with the standard implementation of TriNetX platform. Covariate balance before and after matching was assessed using standardized mean differences, with values below 0.10 considered indicative of adequate balance.

Although PSM achieved adequate balance for most baseline covariates, small residual imbalances remained for selected metabolic and nutritional variables, including hemoglobin A1c, BMI, and serum albumin. Given their known clinical relevance and standardized mean differences exceeding 0.10 after matching, these variables were prespecified for inclusion in subsequent multivariable time-to-event analyses.

Statistical analysis

Baseline characteristics were summarized using means and standard deviations for continuous variables and percentages for categorical variables. Comparisons between cohorts were performed using appropriate statistical tests as implemented within the TriNetX platform.

Time-to-event analyses were performed using Kaplan-Meier survival curves, with differences between cohorts assessed using the log-rank test.

To account for residual imbalance after propensity score matching, a multivariable Cox proportional hazards regression model was constructed, including variables with standardized mean differences greater than 0.10 following matching, specifically hemoglobin A1c, BMI, and serum albumin levels, in addition to the exposure variable (ankle arthrodesis versus

Patients were followed for up to three years from the index surgical procedure. This 3-year follow-up window was selected a priori to ensure uniform outcome ascertainment across the study period and to minimize bias related to differential loss to follow-up in a large, multi-institutional administrative database. This time frame captures most early and intermediate implant-related complications while maintaining consistent censoring across cohorts. Follow-up was censored at the time of implant failure, death, loss to follow-up, or at 3 years after the index date, whichever occurred first. Death was treated as a censoring event and was not explicitly modeled as a competing risk. Although patients were included between January 2012 and December 2023, all time-to-event analyses were anchored to the individual index date, allowing uniform assessment of outcomes within the predefined 3-year follow-up window and appropriate handling of right censoring.

The proportional hazards assumption for the Cox regression model was assessed using graphical inspection of log-log survival plots and evaluation of Schoenfeld residuals. No clinically meaningful violations of the proportional hazards assumption were identified.

Additional multivariable analysis

Given the residual imbalance after propensity score matching, an additional Cox proportional hazards regression model was constructed. This model included covariates with standardized mean differences greater than 0.10 after matching, specifically BMI, hemoglobin A1c, and serum albumin levels. The primary exposure variable was cohort membership (ankle arthrodesis versus total ankle arthroplasty).

Hazard ratios (HRs) with 95% CI were reported. Statistical significance was defined as a two-sided p-value < 0.05.

The statistical programs used were R and SPSS Statistics version 19.0, in addition to four programs or language tools intrinsic to the TriNetX platform.

Ethics

Results

A total of 412,471 patients with a diagnosis of ankle osteoarthritis between 2012 and 2023 were identified. After applying exclusion criteria, 7,973 patients who underwent surgical treatment were included in the final analysis (Figure 1). Of these, 4,825 underwent ankle arthrodesis and 3,148

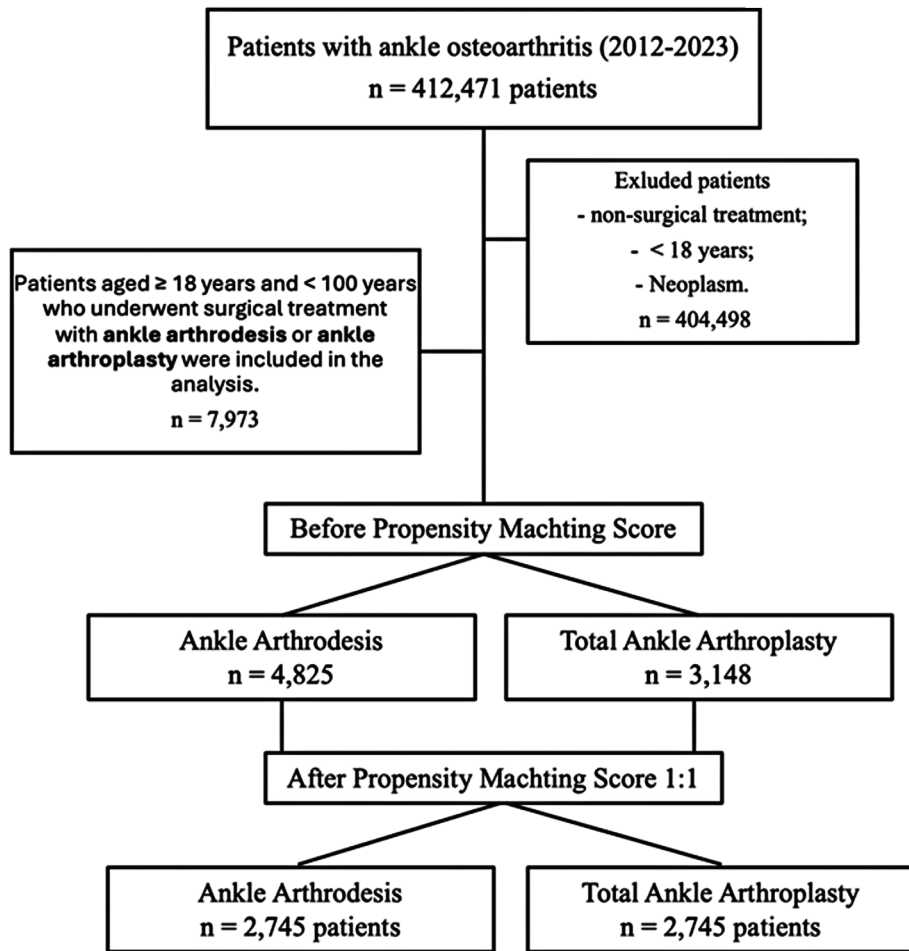


Figure 1. Flow diagram of cohort identification, eligibility, and propensity score matching.

Before propensity score matching, patients in the ankle arthrodesis cohort were younger than those undergoing arthroplasty (mean age, 55.3 ± 14.1 vs 60.9 ± 12.0 years) and had a higher prevalence of several comorbidities, including smoking, diabetes mellitus, obesity, chronic kidney disease, and Charcot arthropathy. Significant baseline differences were also observed in race, ethnicity, prior ankle-related procedures, and laboratory values. Several covariates demonstrated standardized mean differences (SMDs) > 0.10 , indicating substantial imbalance between groups.

Preoperative laboratory data were incompletely available. Hemoglobin A1c values were present for 13.7% and 14.7% of patients in the arthrodesis and arthroplasty groups, respectively, while serum albumin levels were available for 23.4% and 23.8%.

Propensity score matching generated 2,745 well-balanced

between cohorts (59.7 ± 12.6 vs 59.7 ± 12.0 years), and most demographic and clinical variables were adequately balanced (SMDs < 0.10) (Table 1, Figure 2). Small residual imbalances persisted for hemoglobin A1c, serum albumin, and BMI; these variables were subsequently adjusted for in multivariable Cox regression analyses.

All comparative outcome analyses were performed using the matched cohorts.

Outcome

Outcomes analyses were performed in the propensity score-matched cohorts. After exclusion of patients who experienced implant failure prior to the defined time window, 2,074 patients remained in the ankle arthrodesis cohort and 2,249 in the total ankle arthroplasty cohort. A total of 361 implant failures occurred in the ankle arthrodesis group and

ble 1. Baseline demographic and clinical characteristics before and after propensity score matching in patients undergoing ankle arthrodesis and total ankle ar

ID-10	Demographics	Before Propensity Score Matching					After Propensity Score Matching								
		Ankle Arthrodesis (n=4,825)		Total Ankle Arthroplasty (n=3,145)			Ankle Arthrodesis (n=2,745)		Total Ankle Arthroplasty (n=2,745)						
		Mean ± SD	n	% cohort	Mean ± SD	n	% cohort	Std diff.	Mean ± SD	n	% cohort	Mean ± SD	n	% cohort	
	Age at Index	55.3 ± 14.1	4,825	100%	60.9 ± 12.0	3,148	100%	0.430	59.7 ± 12.6	2,745	100%	59.7 ± 12.0	2,745	100%	
	Male		2,757	57.1%		1,701	54%	0.063		1,515	55.2%		1,515	54.7%	
	Female		2,066	42.8%		1,446	45.9%	0.063		1,229	44.8%		1,244	45.3%	
	Race														
	White		3,840	79.6%		2,702	85.8%	0.166		2,349	85.6%		2,340	85.2%	
	Black or African American		460	9.5%		171	5.4%	0.156		167	6.1%		168	6.1%	
	Unknown Race		298	6.2%		106	3.4%	0.132		102	3.7%		104	3.8%	
	Ethnicity														
	Not Hispanic or Latino		3,727	77.2%		2,502	79.5%	0.054		2,188	79.7%		2,169	79.0%	
	Hispanic or Latino		300	6.2%		90	2.9%	0.162		81	3.0%		88	3.2%	
	Unknown Ethnicity		798	16.5%		556	17.7%	0.030		476	17.3%		488	17.8%	
	Comorbidity														
	Smoking		600	12.4%		170	5.4%	0.249		167	6.1%		169	6.1%	
	Alcohol abuse		92	1.9%		23	0.7%	0.103		22	0.8%		22	0.8%	
	Essential (primary) hypertension		1,560	32.3%		939	29.8%	0.054		795	29.0%		799	29.1%	
08-E13	Diabetes mellitus		896	18.6%		252	8.00%	0.315		242	8.8%		251	9.1%	
14.67	Charcot's joint, ankle and foot		314	6.5%		10	0.3%	0.346		10	0.4%		10	0.4%	
4.4	chronic obstructive pulmonary disease		190	3.9%		59	1.9%	0.123		55	2.0%		56	2.0%	
7.2	Venous insufficiency (chronic) (peripheral)		65	1.3%		25	0.8%	0.054		23	0.8%		24	0.9%	
22.322	Carrier or suspected carrier of Methicillin resistant Staphylococcus aureus		34	0.7%		10	0.3%	0.054		10	0.4%		10	0.4%	
18	Chronic kidney disease (CKD)		286	5.9%		80	2.5%	0.169		63	2.3%		73	2.7%	
66	Overweight and obesity		900	18.7%		415	13.2%	0.150		364	13.3%		372	13.6%	
	Body Mass Index (Kg/m2)		3,597	74.5%		2,256	71.7%	0.274		1,964	71.5%		1,994	72.6%	
	> 30 kg/m²		2,359	48.9%		1,306	41.5%	0.149		1,197	43.6%		1,194	43.5%	
	Procedure														
	Surgical Procedures on the Leg (Tibia and Fibula) and Ankle Joint		873	18.1%		218	6.9%	0.343		210	7.7%		213	7.8%	
	Aspiration and/or intra-articular injection of the ankle		455	9.4%		501	15.9%	0.196		336	12.2%		359	13.1%	
	Medications														
	Anabolic steroids		10	0.2%		10	0.3%	0.022		10	0.4%		10	0.4%	
	Corticosteroid for systemic use		1,630	33.8%		1,034	32.8%	0.020		816	29.7%		846	30.8%	
	Fracture														
32.309D	History of closed distal tibial fracture with complete healing.		74	1.5%		10	0.3%	0.127		10	0.4%		10	0.4%	
32.309E	History of an open Gustilo-Anderson type I or II distal tibial fracture, healed		74	1.5%		10	0.3%	0.127		10	0.4%		10	0.4%	
	Laboratory														
	Hb1Ac (mg/dL)		6.5 ± 1.5	10.24	21.2%	5.9 ± 0.9	448	14.2%	0.468	6.2 ± 1.2	375	13.7%	5.9 ± 1.0	403	14.7%
	Serum albumin level (mg/dL)		3.9 ± 0.6	1,434	29.7%	4.2 ± 0.4	767	24.4%	0.480	4.1 ± 0.5	642	23.4%	4.2 ± 0.4	654	23.8%
	< 6.0 mg/dL		1,434	29.7%		767	24.4%	0.121		642	23.4%		654	23.8%	
	Total serum protein level (mg/dL)		71 ± 0.7	1,374	28.5%	71 ± 0.5	727	23.1%	0.006	71 ± 0.6	621	22.6%	71 ± 0.5	616	22.4%
	<3.5 mg/dL		10	0.2%		0	0.0%	0.064		10	0.4%		0	0	0.086

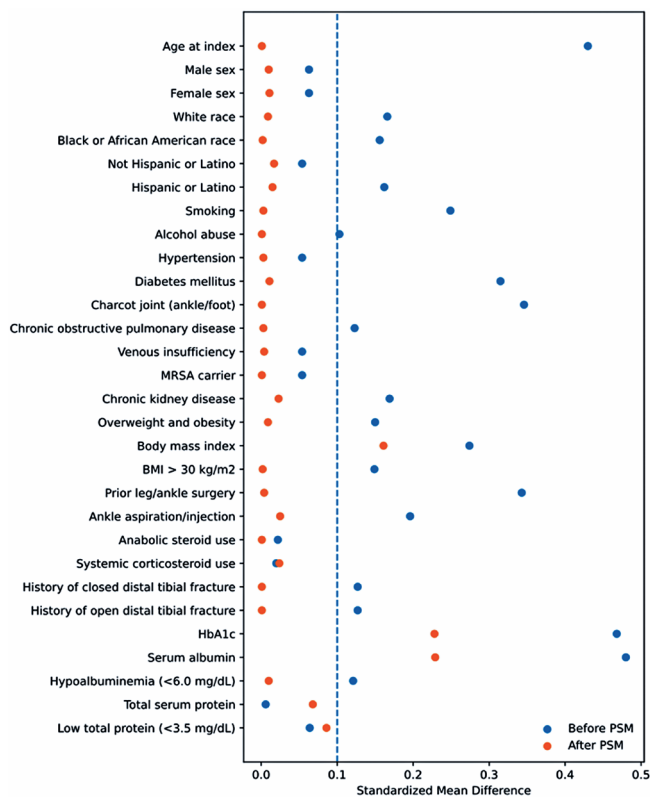


Figure 2. Standardized mean differences demonstrating covariate balance before and after propensity score matching.

Survival analysis

Kaplan-Meier survival analysis demonstrated significantly lower implant survival in the ankle arthrodesis cohort over the follow-up period. Survival probability at the end of the observation window was 77.9% for ankle arthrodesis and 84.5% for total ankle arthroplasty. The difference between groups was statistically significant by log-rank testing ($\chi^2 = 29.48$; $p < 0.001$). Median survival was not reached in either cohort during the study period (Figure 3).

Cox proportional hazards analysis

This model was specifically designed to adjust for residual metabolic and nutritional imbalances observed after propensity score matching.

In multivariable Cox proportional hazards analysis, adjusting for residual metabolic and nutritional imbalances, ankle arthrodesis remained independently associated with an increased hazard of implant failure compared with total ankle arthroplasty (HR, 1.94; 95% CI, 1.75-2.15; $p < 0.001$). Elevated hemoglobin A1c was also independently associated with an increased risk of implant failure (HR, 1.36; 95% CI, 1.21-1.54; $p < 0.001$). Serum albumin level (HR, 1.10; 95% CI, 0.99-1.22; $p = 0.074$) and BMI (HR, 1.08; 95% CI, 0.97-1.22; $p = 0.150$) were not independently associated with implant failure after adjustment (Table 3).

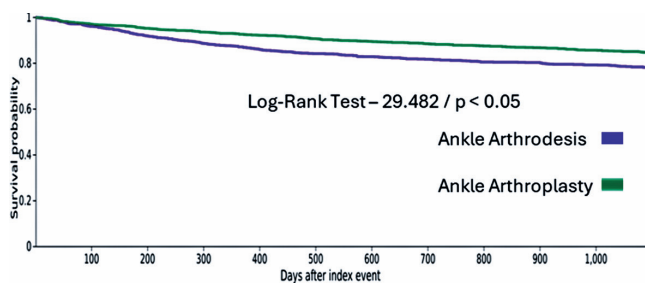


Figure 3. Kaplan-Meier survival curve for implant failure comparing ankle arthrodesis and total ankle arthroplasty.

Table 2. Association for implant failure after propensity score matching

Cohort	Cohort statistic		
	Patients in cohort	Patients with outcome	Risk
1 Ankle Arthrodesis	2,074	361	17,4%
2 Total Ankle Arthroplasty	2,249	271	12%
Risk difference	Risk difference		
	95% CI	z	p
Risk ratio	Risk ratio		
	95% CI	Odds ratio	
		Odds ratio	95% CI
5.4%	(0.032-0.075)	4.98	0.0
		1.445	(1.249-1.671)
		1.538	(1.297-1.824)

proportional hazards regression analysis evaluating factors associated with implant failure after ankle arthroplasty

	Hazard ratio	Coefficient	Standard error	z	P > z	95% CI
Arthrodesis or Arthroplasty	1.944	0.665	0.053	12.586	0.000	(1.753-2.156)
Hemoglobin A1c/	1.369	0.314	0.060	5.253	0.000	(1.218-1.540)
Albumin (mg/dL)	1.103	0.098	0.055	1.787	0.074	(0.991-1.229)
BMI (kg/m ²)	1.089	0.085	0.059	1.439	0.150	(0.970-1.223)

BMI: Body mass index; CI: Confidence interval.

Discussion

In this retrospective comparative cohort study using the international TriNetX database, ankle arthrodesis was associated with a significantly higher risk of coded implant failure than total ankle arthroplasty. This association remained consistent across multiple analytic approaches, including propensity score matching and adjusted time-to-event analyses.

A key consideration when interpreting these findings is the conceptual non-equivalence of the outcome between procedures. Although both involve implanted materials, administrative implant failure codes capture different underlying mechanisms. In arthrodesis, these codes often reflect nonunion-related hardware complications or symptomatic fixation, whereas in arthroplasty, they more commonly represent prosthesis-specific failures, such as loosening, wear, or infection. Consequently, the outcome reflects heterogeneous implant-related complications rather than directly comparable biological or mechanical failures and may systematically disadvantage the arthrodesis cohort.

Historically, ankle arthrodesis has been regarded as a durable and reliable treatment for end-stage ankle osteoarthritis, particularly among younger patients and those with greater comorbidity burdens⁽¹¹⁾. Early comparative studies suggested similar intermediate-term survivorship between arthrodesis and arthroplasty, but were limited by small sample sizes, heterogeneous implant designs, and non-standardized selection criteria⁽¹²⁻¹⁴⁾. Haddad et al.⁽¹⁵⁾ emphasized these methodological challenges in their systematic review. With modern implant designs and improved surgical techniques, more recent prospective and multicenter investigations have demonstrated progressively better outcomes following total ankle arthroplasty^(16,17). Saltzman et al.⁽¹⁸⁾ reported acceptable early survivorship of the STAR prosthesis, and Daniels et al.⁽¹⁹⁾, in the COFAS multicenter study, observed comparable survival with superior functional outcomes for arthroplasty. Similar advantages were described by Glazebrook et al.⁽²⁰⁾.

The present study extends this literature by leveraging a large, contemporary population and applying propensity score matching to reduce indication bias. These real-world data suggest a higher rate of coded implant-related complications following arthrodesis compared with arthroplasty.

Our findings are consistent with national registry and meta-

ankle arthroplasty. Henricson et al.⁽²¹⁾ reported favorable long-term survival in registry analyses, while Zaidi et al.⁽²²⁾ showed progressive improvements in arthroplasty outcomes over time. Functional and gait studies further suggest that biomechanics are more physiological and that quality of life is better after arthroplasty compared with fusion^(23,24).

Several limitations warrant consideration. This analysis relies on administrative ICD-10 and CPT coding, which limits clinical detail and introduces potential misclassification. Important operative variables—including implant type, fixation method, surgical technique, deformity severity, bone quality, and surgeon experience—were unavailable and may contribute to residual confounding. In addition, the ICD-based implant failure definition aggregates heterogeneous complications that are not directly comparable between procedures and may overestimate failure after arthrodesis. Laboratory data were incomplete, restricting some analyses to complete cases and potentially introducing selection bias. Although longer-term outcomes at five or ten years are clinically relevant, extended follow-up in administrative databases is often affected by incomplete longitudinal capture and differential attrition. A 3-year horizon was therefore selected to balance clinical relevance with methodological robustness. Finally, death was treated as a censoring event rather than explicitly modeled as a competing risk, which may bias estimates if mortality differs between groups.

Despite these limitations, this study is strengthened by its large sample size, multicenter international data source, contemporary timeframe, and adjustment for key metabolic and nutritional factors. The consistency of results across analytic strategies supports the robustness of the observed association.

Conclusion

In this large propensity score-matched analysis of patients with ankle osteoarthritis, ankle arthrodesis was associated with a significantly higher risk of coded implant failure compared with total ankle arthroplasty. These findings suggest that, when patient-specific factors permit, total ankle arthroplasty may represent a durable alternative to arthrodesis in appropriately selected patients. Future prospective studies incorporating detailed radiographic parameters, implant-specific data, and patient-reported outcomes are needed to further refine surgical decision-making and optimize patient

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